FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST - IV
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	19.	**B**	37.	**A**
2.	**C**	20.	**A**	38.	**B**
3.	**B**	21.	**A**	39.	**C**
4.	**A**	22.	**C**	40.	**D**
5.	**A**	23.	**A**	41.	**A**
6.	**ABC**	24.	**ACD**	42.	**C**
7.	**AD**	25.	**BCD**	43.	**ACD**
8.	**ACD**	26.	**BCD**	44.	**ABCD**
9.	**AD**	27.	**ABCD**	45.	**AD**
10.	**ACD**	28.	**AC**	46.	**BD**
11.	**ACD**	29.	**ABCD**	47.	**AB**
12.	**ABD**	30.	**BD**	48.	**ABC**
13.	**BC**	31.	**ABCD**	49.	**ABC**
14.	**5**	32.	**5**	50.	**8**
15.	**6**	33.	**7**	51.	**5**
16.	**8**	34.	**9**	52.	**3**
17.	**1**	35.	**3**	53.	**9**
18.	**6**	36.	**2**	54.	**0**

Physics

PART – I

SECTION – A

1. D
2. C

From the equation of trajectory $y = 10\tan\theta - \frac{20}{9}\sec^2\theta$

$\frac{dy}{d\theta} = 0$ gives, $\tan\theta = \left(\frac{9}{4}\right)$

3. B

$4 \times 2\pi = \frac{1}{2} \times \alpha \times 1^2, \quad \alpha = 16\pi$

Also, $F = \frac{mr\alpha}{2}$

4. A

Applying conservation of angular momentum about Q

$mvR\sin 30 = \left(\frac{mvd}{\sqrt{3}}\right)$

$\therefore d = \frac{\sqrt{3}}{2}R$

5. A

6. ABC

7. AD

8. ACD

9. AD

10. ACD

11. ACD

12. ABD

13. BC

m, M will separate when a_x for 'm' becomes zero.

SECTION – C

14. [5]

15. [6]

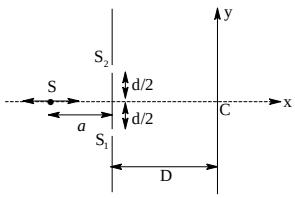
16. [8]

17. [1]

In the CM frame, work done by pseudo forces $= 2\left(\frac{F}{2}\times \ell\right) = F\ell$, this is the energy lost in collision.

18. [6]

$$I = \frac{P}{4\pi r^2}$$



$$I = \frac{P}{4\pi\left(a^2 + \left(\frac{d}{2}\right)^2\right)} \Rightarrow I_{\max} = 4I$$

Differentiate to find answer

Chemistry

PART – II

SECTION – A

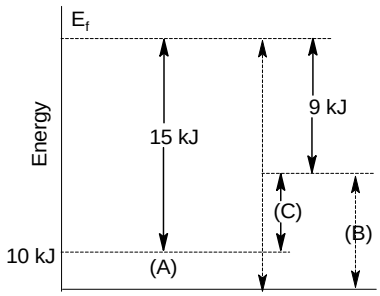
19. B
20. A
21. A
22. C
23. A
Use trigonometry formulas
24. ACD
25. BCD
26. BCD
27. ABCD
28. AC

Sol: $p^H = 2 \Rightarrow [H^+] = 10^{-2}$ $[Ca^{+2}] = \frac{10^{-2}}{2} = 5\times10^{-3}$ moles.

$= 5\times10^{-3}$ x 40 = 200 mg.

X = 200 ppm.

26. ABCD



Hence, (A), (B), (C) and (D) are correct answer.

30. BD
31. ABCD

SECTION – C

32. [5]
33. [7]
D is wrong other statements are correct
This ore is $FeCr_2O_4$ and it is the prep. Of $K_2Cr_2O_7$ in figure
$FeCr_2O_4 + O_2 + Na_2CO_3 \rightarrow Na_2CrO_4(\text{yellow}) + Fe_2O_3 + CO_2$ after acidification we have
$NaCr_2O_7.2H_2O$ and $Na_2SO_4.10H_2O$
34. [9]
Except chloric acid all have
35. [3]
$V_2O_5, V_2O_3, CrO, Cr_2O_3, Mn_2O_7, ZnO$
(an) (b) (b) (an) (a) (an)
36. [2]

Mathematics

PART – III

SECTION – A

37. A

$$x+\sqrt{x^2+1}=\frac{1}{y+\sqrt{y^2+1}}$$

$$\ell n(x+\sqrt{x^2+1})=\ell n\left(\frac{1}{y+\sqrt{y^2+1}}\right)$$

$$x+y=\sqrt{x^2+1}+\sqrt{y^2+1}.....(1)$$

Similarly, $y+\sqrt{y^2+1}=\frac{1}{x+\sqrt{x^2+1}}$

$x+y=-\sqrt{x^2+1}-\sqrt{y^2+1}.....(2)$ ading1and 2 x+y=0

Image of xy=1in xy=1 itself

38. B

$\frac{1}{r_1},\frac{1}{r_2},\frac{1}{r_3}$ are in A.P.

$\frac{(s-a)}{\Delta},\frac{(s-b)}{\Delta},\frac{(s-c)}{\Delta}$ are in A.P.

A, b, c in A.P.

39. C

$|A|^{(n-1)^2}=|A|^{(2n^2-7x+7)}$

$n^2-2n+1=2n^2-7n+7$

$n^2-5n+6=0$

$\Rightarrow n=3,2$

40. D

Let $I=\int_0^3\frac{f(x)dx}{(x^2-3x+1)}.....(1)$

$I=\int_0^3\frac{f(3-x)}{(x^2-3x+1)}.....(2)$

$(1)+(2)\Rightarrow 2I=3$

$I=\frac{3}{2}$

41. A

$y=e^{\frac{3\sin^{-1}x}{4}+\frac{\pi}{8}}$

Maximum when $\sin^{-1}x=\frac{\pi}{2}$

Minimum when $\sin^{-1}x=-\frac{\pi}{2}$

$y=e^{\frac{-3\pi}{8}+\frac{\pi}{8}}=e^{-\frac{\pi}{2}}$

42. C

$V_n=2a_{n+2}+S_n$

$=2[-4+(n+1)]+\frac{n}{2}[-8+(n-1)]$

$=\frac{n^2-5n-12}{2}$

V_n is minimum at n=2 or n=3

$\Rightarrow$ Minimum value of V_n is -9

43. ACD

$8^3x^9+9.8^2x^6+27.8x^3-x+219=0$

$[8x^3+3]^3=512x-192$

$x^3+\frac{3}{8}=\left(x-\frac{3}{8}\right)^{1/3}$

$\Rightarrow f(x)=f^{-1}(x)$

$\Rightarrow f(x)=x$[$\because$ f(x)is continuous and $\uparrow$ fn]

$\Rightarrow x^3+\frac{3}{8}=x$

$x=\frac{1}{2},\frac{-1+\sqrt{13}}{4},\frac{-1-\sqrt{13}}{4}$

44. ABCD

$x^3+2x^2-3x+1=0\begin{cases}\alpha\\ \beta\\ \gamma\end{cases}$

$x^3-3x^2-2x+1=0\begin{cases}1/\alpha\\ 1/\beta\\ 1/\gamma\end{cases}$

Given expression $\frac{1}{3-\frac{1}{\alpha}}+\frac{1}{3-\frac{1}{\beta}}+\frac{1}{3-\frac{1}{\gamma}}$

$=\frac{\sum\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)}=\frac{27-6\sum\left[\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right]+\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)}=\frac{11}{7}$

45. AD

Given $f(f(x))=x$

$\Rightarrow \frac{af(x)+b}{cf(x)+d}=x$

$\Rightarrow f(x)=\frac{-dx+b}{cx-a}$.....(1)

Given $f(x)=\frac{ax+b}{cx+d}$.....(2)

from(1)&(2)a=-d.....(3)

f(19)=19

$\Rightarrow$ 38a+b=97×97c.....(4)

f(97)=97

$\Rightarrow 194a+b=97\times 97$c.....(5)

from (5)-(4)

a=58c

$\therefore \alpha=58=2^1.29^1$

46. BD

$f(a_1)\le f(a_2)\le f(a_3)\le .....\le f(a_{100})$ It is equivalent to distribution of 100 objects to 50 persons such that each person get atleast one object.

$\therefore$ number of such mapings= $^{99}C_{49}=^{99}C_{50}$

47. AB

x^3-x^2+bx+c=0.....(1)

Let the roots of the equation be a-d,a+d

Sum of the roots=1

a-d+a+a+d=1 or a=1/3.....(2)

(a-d)a+a(a+d)+(a-d)(a+d)=b

or $3a^2-d^2=b$

or $3(1/9)-d^2=b$ or $1/3-d^2=b$

or $d^2=1/3-b$

Since d is real, 1/3-b>=0

b<=1/3

thus <b<=1/3

Product of roots=-c

(a-d)a(a+d)=-c

$a(a^2-d^2)=-c$

$1/3(a/9-d^2)=-c$

as d is real

$d^2>=0$

Or $\Rightarrow c\ge -\frac{1}{27}\Rightarrow -\frac{1}{27}\le c<\infty$

48. ABC

$$3x^2\frac{dx}{dy}=\frac{x^3}{x^3-y} \qquad x^3=t$$

$$\Rightarrow \frac{dt}{dy}=\frac{t}{t-y} \quad \Rightarrow (t-y)dt=t\,dy$$

$$\Rightarrow t\,dt=y\,dt+t\,dt \Rightarrow \frac{t^2}{2}=t.y+\frac{c}{2}$$

$$\Rightarrow x^6=2x^3y+c$$

49. ABC

the 7-digit Numbers with 1 in the left most place and containing each of the digits $1,2,3,4,5,6,7$ exactly once is $6!=720$.

SECTION – C

50. [8]

The triangle has circumcentre at origin and its orthocentre laying on the circumcircle.

51. [5]

$$\lim_{y\to 1}\frac{100y^{100}}{y^{100}-1}-\frac{50.y^{50}}{y^{50}-1}$$

$$\lim_{y\to 1}100-50-\left(\frac{100}{1-y^{100}}-\frac{50}{1-y^{50}}\right)$$

$$\text{L=50-L} \therefore \text{L=}\frac{50}{2}=25$$

52. [3]

No condition on I, j, k

$$\frac{3}{2}\times\frac{3}{2}\times\frac{3}{2}=\frac{27}{8}$$

When any two of I, j, k are equal

$$\sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{1}{3^{2i}}\cdot\frac{1}{3^{j}}=\frac{3}{2}\left[1+\frac{1}{9}+.....\right]$$

$$\frac{3}{2}\times\frac{9}{8}=\frac{27}{16}$$

Where all three equal $\frac{27}{16}$

Required $\frac{27}{8}-3.\frac{27}{16}+\frac{2.27}{26}=\frac{81}{208}$

53. [9]

We observe that

$$f(9)=f(4+5)=f(4.5)=f(16+4)$$

$$=f(16.4)=f(64)=f(8.8)$$

$$=f(8+8)=f(16)=f(4.4)=f(4+4)=f(8)=9$$

54. [0]

Let $x=\alpha$ be common zero then

$\alpha^2+a\alpha+b=0.....(1)$

$\alpha^2+\alpha+ab=0.....(2)$

$a\alpha^2+\alpha+b=0.....(3)$

$(1)-(2) \Rightarrow \alpha=b$ put in

(1) or (3)gives $ab^2+ab+b=0.....(4)$

& (3) give $ab^2+2b=0$ $b \neq 0 \Rightarrow ab=-2$

$a=-2, b=1 and\ a=1, b=-2$ (does not satisfy the mentioned condition polynomial will be same) $a+2b=0$.